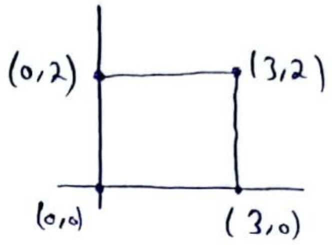
Q:- Find CoM & Radii of gyration for a lamina having mass density of $\rho(x,y) = y$ & occupying the region $(0,0)$, $(3,0)$, $(3,2)$, $(0,2)$. Also sketch the region. (a)

C1 CoM
C2 RoG

Sol



$$\text{mass} = M = \int_0^2 \int_0^3 y \, dx dy$$

$$= \int_0^2 y \, x\Big|_0^3 dy$$

$$= \int_0^2 3y \, dy$$

$$= \frac{3y^2}{2}\Big|_0^2$$

$$= 6$$

$$M_x = \int_0^2 \int_0^3 y \cdot y \, dx dy$$

$$= \int_0^2 \int_0^3 y^2 \, dx dy$$

$$= \int_0^2 y^2 \, x\Big|_0^3 dy$$

$$= \int_0^2 3y^2 dy$$

$$= y^3\Big|_0^2$$

$$= 8$$

$$M_y = \int_0^2 \int_0^3 xy \, dx dy$$

$$= \int_0^2 \frac{1}{2} y \cdot x^2\Big|_0^3 dy$$

$$= \int_0^2 \frac{9}{2} y \, dy$$

$$= \frac{9}{4} y^2 \Big|_0^2$$

$$= 9$$

$$\bar{x} = \frac{M_y}{m}$$

$$\bar{x} = \frac{9}{6} = \frac{3}{2}$$

$$\bar{y} = \frac{M_x}{m}$$

$$\bar{y} = \frac{8}{6} = \frac{4}{3}$$

$$CoM = (\bar{x}, \bar{y}) = \left(\frac{3}{2}, \frac{4}{3}\right)$$

Moment of Inertia about x-axis

$$I_x = \int_0^2 \int_0^3 y^2 \cdot y \, dx \, dy$$

$$= \int_0^2 y^3 x \Big|_0^3 dy$$

$$= \int_0^2 3y^3 \, dy$$

$$= \frac{3}{4} y^4 \Big|_0^2$$

$$= 12$$

$$I_y = \int_0^2 \int_0^3 x^2 \cdot y \, dx \, dy$$

$$= \int_0^2 \frac{1}{3} y \cdot x^3 \Big|_0^3 dy$$

$$= \int_0^2 9y \, dy$$

$$= \frac{9}{2} y^2 \Big|_0^2$$

$$= 18$$

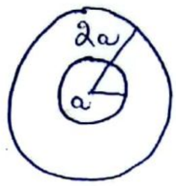
$$I_o = 12 + 18 = 30$$

B2

Q:- Find the mass of a flat circular washer with inner radius 'a' & outer radius '2a' if its mass density $\rho = \rho(x,y)$ is inversely proportional to the square of the distance from the center.

Sol

$$\rho(x,y) = \frac{K}{x^2+y^2}$$



$$\text{Mass} = \iint_D \rho(x,y) \, dA$$

$$= \int_0^{2\pi} \int_a^{2a} \frac{K}{\cancel{x^2+y^2}\; r^2} \, r \, dr \, d\theta$$

$$= \int_0^{2\pi} K \ln r \Big|_a^{2a} \, d\theta$$

$$= \int_0^{2\pi} K \left(\ln(2a) - \ln(a) \right) d\theta$$

$$= K \left[\ln(2a) - \ln(a) \right] \cdot \left[\theta \right]_0^{2\pi}$$

$$= K \left[\ln\left(\frac{2a}{a} \right) \right] \cdot (2\pi)$$

$$= 2\pi K \ln 2$$

C1

Q:- Find out moment of inertia I_x, I_y & I_o of a homogeneous disk D with density $\rho(x,y) = \rho = 1$, center the origin & radius 'a'. (using P.C's also specify limits)

Sol Boundary of D is Circle $x^2+y^2=a^2$ & in polar Coordinates D is described by $0 \le \theta \le 2\pi$, $0 \le r \le a$.

$$I_x = \iint_D y^2 \rho \, dA$$

$$= \rho \int_0^{2\pi}\int_0^a (r\sin\theta)^2 r\,dr\,d\theta$$

$$= \rho \int_0^{2\pi}\int_0^a r^3 \sin^2\theta \, dr\,d\theta$$

$$= \rho \int_0^{2\pi} \sin^2\theta \cdot \left[\frac{r^4}{4}\right]_0^a d\theta$$

$$= \rho \int_0^{2\pi} \sin^2\theta \cdot \frac{a^4}{4} \, d\theta$$

$$= \frac{\rho a^4}{4} \int_0^{2\pi} \left(\frac{1-\cos 2\theta}{2}\right) d\theta$$

$$= \frac{\rho a^4}{4} \left[\frac{1}{2}\theta - \frac{\sin 2\theta}{4}\right]_0^{2\pi}$$

$$= \frac{\rho a^4 \pi}{4}$$ if $\rho = 1$ then $\frac{a^4\pi}{4}$

$$I_y = \iint_D x^2 \rho \, dA$$

$$= \rho \int_0^{2\pi}\int_0^a (r\cos\theta)^2 r\,dr\,d\theta$$

$$= \rho \int_0^{2\pi}\int_0^a r^3 \cos^2\theta \, dr\,d\theta$$

$$= \rho \int_0^{2\pi} \cos^2\theta \left[\frac{r^4}{4}\right]_0^a d\theta$$

$$= \rho \int_0^{2\pi} \cos^2\theta \cdot \frac{a^4}{4}\, d\theta$$

$$= \frac{\rho a^4}{4} \int_0^{2\pi} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= \frac{\rho a^4}{4} \left[\frac{1}{2}\theta + \frac{\sin 2\theta}{4}\right]_0^{2\pi}$$

$$= \frac{\pi \rho a^4}{4}$$ (Symmetry) If $\rho = 1$ $\frac{\pi a^4}{4}$

$$I_o = I_x + I_y$$

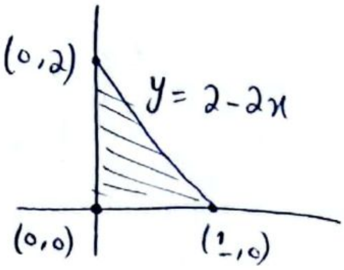
$$I_o = \frac{\pi \rho a^4}{4} + \frac{\pi \rho a^4}{4}$$ If $\rho = 1$ then $\frac{\pi a^4}{2}$

$$I_o = \frac{\pi \rho a^4}{2}$$

Q:- Find mass & CoM of a triangular lamina with vertices $(0,0)$, $(1,0)$ & $(0,2)$ if density fn is $\rho(x,y) = 1+3x+y$ also sketch region.

Sol

$$m = \int_0^1 \int_0^{2-2x} (1+3x+y)\, dy\, dx$$



$$= \int_0^1 \left[y + 3xy + \frac{y^2}{2}\right]_0^{2-2x} dx$$

$$= \int_0^1 \left((2-2x) + 3x(2-2x) + \frac{(2-2x)^2}{2}\right) dx$$

$$= \int_0^1 \left(2-2x+6x-6x^2 + \frac{4-8x+4x^2}{2}\right) dx$$

$$= \int_0^1 \frac{4-4x+12x-12x^2+4-8x+4x^2}{2}\, dx$$

$$= \int_0^1 \frac{8-8x^2}{2}\, dx$$

$$= \int_0^1 \frac{8(1-x^2)}{2}\, dx$$

$$= 4\int_0^1 (1-x^2)\, dx$$

$$= 4\left[x-\frac{x^3}{3}\right]_0^1$$

$$= \frac{8}{3}$$

$$M_x = \int_0^1\int_0^{2-2x} y\,(1+3x+y)\, dy\, dx$$

$$= \int_0^1\int_0^{2-2x} (y+3xy+y^2)\, dy\, dx$$

$$= \int_0^1 \left[\frac{y^2}{2}+\frac{3xy^2}{2}+\frac{y^3}{3}\right]_0^{2-2x} dx$$

$$= \int_0^1 \left[\frac{(2-2x)^2}{2}+\frac{3x(2-2x)^2}{2}+\frac{(2-2x)^3}{3}\right] dx$$

$$= \int_0^1 \left(\frac{4-8x+4x^2}{2}+\frac{12x-24x^2+12x^3}{2}+\frac{8-24x+24x^2-8x^3}{3}\right) dx$$

$$= \int_0^1 \left(\frac{12 - 24x + 12x^2 + 36x - 72x^2 + 36x^3 + 16 - 48x + 48x^2 - 16x^3}{6} \right) dx$$

$$= \int_0^1 \frac{28 - 36x - 12x^2 + 20x^3}{6} \, dx$$

$$= \int_0^1 \frac{2}{6} \left(14 - 18x - 6x^2 + 10x^3 \right) dx$$

$$= \frac{1}{3} \left[14x - \frac{18x^2}{2} - \frac{6x^3}{3} + \frac{10x^4}{4} \right]_0^1$$

$$= \frac{1}{3} \left[14 - \frac{18}{2} - \frac{6}{3} + \frac{10}{4} \right]$$

$$= \frac{1}{3} \left[\frac{11}{2} \right]$$

$$= \frac{11}{6}$$

$$\bar{y} = \frac{11/6}{8/3} = \frac{11}{16}$$

$$M_y = \int_0^1 \int_0^{2-2x} x \left(1 + 3x + y \right) dy \, dx$$

$$= \int_0^1 \int_0^{2-2x} \left(x + 3x^2 + xy \right) dy \, dx$$

$$= \int_0^1 \left[xy + 3x^2 y + \frac{xy^2}{2} \right]_0^{2-2x} dx$$

$$= \int_0^1 \left[x(2-2x) + 3x^2(2-2x) + \frac{x(2-2x)^2}{2} \right] dx$$

$$= \int_0^1 \left(2x - 2x^2 + 6x^2 - 6x^3 + \frac{4x - 8x^2 + 4x^3}{2}\right) dx$$

$$= \int_0^1 \left(\frac{4x - 4x^2 + 12x^2 - 12x^3 + 4x - 8x^2 + 4x^3}{2}\right) dx$$

$$= \int_0^1 \left(\frac{8x - 8x^3}{2}\right) dx$$

$$= \int_0^1 \frac{8(x - x^3)}{2} dx$$

$$= 4 \int_0^1 (x - x^3)\, dx$$

$$= 4 \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1$$

$$= 4 \left[\frac{1}{2} - \frac{1}{4}\right]$$

$$= 4 \cdot \frac{1}{4} = 1$$

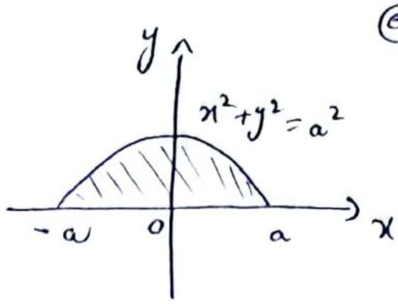
$$\bar{x} = \frac{3}{8}$$

CoM is at $\left(\frac{3}{8}, \frac{11}{16}\right)$

Q:- The density at any pt on^{C2} a semicircular lamina is proportional to the distance from center of circle. If both lamina & density ftn are symmetric wrt y-axis & CoM lie on y-axis, that is, $\bar{x} = 0$. Then find $\bar{y}$ using P.C's & also sketch region. Take radius 'a'.

Sol

$\rho(x,y) = k\sqrt{x^2+y^2}$ (e)



$0 \le r \le a$

$0 \le \theta \le \pi$

$$m = \int_0^{\pi}\int_0^{a} (kr)\, r\, dr\, d\theta$$

$$m = \int_0^{\pi} \frac{kr^3}{3}\Big|_0^a \, d\theta$$

$$= \int_0^{\pi} \frac{ka^3}{3\cancel{2}}\, d\theta$$

$$= \frac{ka^3}{\cancel{2}3}\, \theta \Big|_0^{\pi}$$

$$= \frac{ka^3\pi}{3}$$

$$\bar{y} = \frac{M_x}{m}$$

$$M_x = \int_0^{\pi}\int_0^{a} (r\sin\theta)(kr)\, r\, dr\, d\theta \qquad \text{As } y = r\sin\theta$$

$$= \int_0^{\pi}\int_0^{a} kr^3 \sin\theta\, dr\, d\theta$$

$$= \int_0^{\pi} k\sin\theta \, \frac{r^4}{4}\Big|_0^a \, d\theta$$

$$= \int_0^{\pi} \frac{k\sin\theta\, a^4}{4}\, d\theta$$

$$= \frac{Ka^4}{4} \int_0^{\pi} \sin\alpha \, d\alpha$$

$$= \frac{Ka^4}{4} (-\cos\alpha)\Big|_0^{\pi}$$

$$= \frac{2Ka^4}{4}$$

$$= \frac{Ka^4}{2}$$

$$\bar{y} = \frac{\frac{Ka^4}{2}}{\frac{Ka^3\pi}{3}}$$

$$= \frac{Ka^4}{2} \times \frac{3}{Ka^3\pi}$$

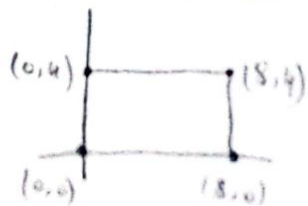
$$= \frac{3a}{2\pi}$$

So CoM is at pt $\left(0, \frac{3a}{2\pi}\right)$

Q:- A town is the shape of a rectangle, with vertices (0,0), (8,0), (8,4) & (0,4). The population density is modeled by the ftn $(x,y) = xy^2$. They want to place their city hall so that it is centered relative to their population, not necessarily geographically. Determine the population center of this town. (Assume the units are miles, & the density is hundreds of people per sq mile). Also sketch region.

Sol $M = \int_0^4 \int_0^8 xy^2 \, dx \, dy$

$$= \int_0^4 y^2 \, dy \int_0^8 x \, dx = \frac{2048}{3} = 682.66$$



$$M_x = \int_0^4 \int_0^8 x y^3 \, dx \, dy$$

$$= \int_0^4 y^3 \, dy \int_0^8 x \, dx$$

$$= 2048$$

$$M_y = \int_0^4 \int_0^8 x^2 y^2 \, dx \, dy$$

$$= \int_0^4 y^2 \, dy \int_0^8 x^2 \, dx$$

$$= \frac{32768}{9}$$

$$\bar{x} = \frac{M_y}{M} = \frac{16}{3} = 5.33$$

$$\bar{y} = \frac{M_x}{M} = 3$$

B1

Q:- A flat metal plate lying in the xy-plane is heated in such a way that the temp at the pt (x,y) in $T^\circ C$, where

$$T(x,y) = 10 y e^{-xy}$$

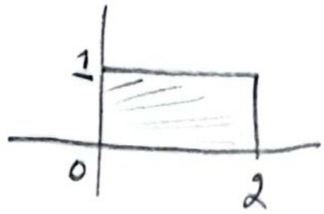
Find the average temp over a rectangular portion of the plate for which $0 \le x \le 2$ & $0 \le y \le 1$.

Sol

$$A = bh$$

$$A = (2-0)(1-0)$$

$$A = 2$$



$$\text{Ave Value} = \frac{1}{A} \iint_R T(x,y)\,dx\,dy$$

$$= \frac{1}{2} \int_0^1 \int_0^2 10 y e^{-xy}\,dx\,dy$$

$$= 1 \cdot 5 \int_0^1 \int_0^2 y e^{-xy}\,dx\,dy$$

$$= 5 \int_0^1 y\left(-\frac{1}{y}\right)\left[e^{-xy}\right]_0^2 dy$$

$$= -5 \int_0^1 \left(e^{-2y} - e^{0}\right) dy$$

$$= -5 \int_0^1 \left(e^{-2y} - 1\right) dy$$

$$= -5 \left[-\frac{1}{2} e^{-2y} - y\right]_0^1$$

$$= -5 \left[\left(-\frac{1}{2} e^{-2} - 1\right) - \left(-\frac{1}{2} e^{0}\right)\right]$$

$$= -5 \left(-\frac{1}{2} e^{-2} - \frac{1}{2}\right) \Rightarrow \frac{5}{2}\left(e^{-2} + 1\right) \approx 2.84^\circ C$$